

RAMAGUNDAM, India

WMO: 430860

Lat: 18.767N

Lon: 79.433E

Elev: 512

StdP: 14.43

Time Zone: 5.50 (IND)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	55.9	58.0	39.3	36.0	82.2	43.7	42.8	80.8	7.0	80.1	5.6	80.5	0.0	230	0.321

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	20.5	109.0	77.5	106.8	77.0	104.6	77.1	83.6	97.7	82.4	95.6	81.4	93.4	2.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.6	161.9	89.0	79.5	156.2	87.7	78.7	151.8	86.8	48.3	98.2	46.8	96.2	45.7	92.9	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.3	7.1	5.7	DB	51.4	112.4	1.9	2.4	50.0	114.1	48.9	115.4	47.9	116.8	46.5	118.5
			WB	48.0	85.2	2.2	0.9	46.4	85.8	45.1	86.3	43.8	86.8	42.2	87.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.1	73.4	79.3	85.2	92.0	95.7	90.1	83.9	82.7	83.4	81.8	76.8	73.0
	DBStd	7.66	3.52	3.62	3.69	3.35	3.97	5.90	3.48	2.81	2.26	2.68	3.29	3.46
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12086	726	821	1092	1259	1418	1204	1049	1012	1001	987	805	712
	CDD65	6611	261	401	627	809	953	754	584	547	551	522	355	247
	CDH74	83065	2110	4011	8012	12437	15741	11307	6932	6075	6249	5307	2959	1925
	CDH80	44978	672	1935	4511	8223	11303	7109	2922	2148	2372	2170	1062	551
(14) Wind	WSAvg	1.6	1.3	1.9	2.0	2.7	2.4	2.4	1.8	1.6	1.2	0.8	0.9	0.7
(15) Precipitation	PrecAvg	38.3	0.5	0.4	0.5	0.6	0.6	5.8	10.0	9.7	6.6	3.0	0.5	0.2
	PrecMax	57.8	2.9	2.7	4.8	3.6	4.5	10.9	18.2	24.1	15.8	9.6	4.4	1.5
	PrecMin	23.3	0.0	0.0	0.0	0.0	0.0	0.9	2.6	2.6	0.3	0.0	0.0	0.0
	PrecStd	9.5	0.9	0.6	1.1	0.9	1.0	2.9	4.7	5.3	4.2	2.6	1.1	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.9	97.6	103.1	109.0	112.7	110.4	96.1	93.9	95.3	94.9	91.1	89.2
		MCWB	68.4	69.5	72.4	76.3	77.2	79.0	77.9	78.8	78.7	75.7	72.7	68.9
	2%	DB	88.0	94.9	100.7	106.7	110.2	106.8	94.1	91.6	92.6	92.7	89.2	86.9
		MCWB	67.9	69.9	73.4	75.4	77.4	78.5	77.8	78.5	78.4	76.2	71.8	68.1
	5%	DB	86.0	92.6	98.6	104.7	108.2	103.7	92.4	90.0	90.7	91.0	87.7	85.2
		MCWB	67.0	69.9	73.3	75.8	77.2	78.1	77.7	78.2	78.1	75.5	70.9	67.3
	10%	DB	84.0	90.3	96.3	102.5	106.1	100.3	90.4	88.4	89.1	89.3	86.0	83.3
		MCWB	66.4	69.2	72.5	75.9	77.5	77.7	77.8	78.0	78.0	74.9	70.1	66.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.4	77.3	81.1	84.3	85.1	84.4	82.4	81.9	81.9	81.8	78.2	74.7
		MCDB	82.8	86.5	91.8	97.4	97.7	98.0	89.6	88.4	88.9	89.8	85.8	82.6
	2%	WB	72.4	74.6	79.0	82.5	84.0	82.8	81.0	80.9	81.1	80.0	76.8	72.4
		MCDB	80.9	86.9	91.2	95.8	99.8	96.5	88.4	87.3	88.1	87.7	84.7	80.6
	5%	WB	70.9	72.8	77.4	80.8	82.7	81.6	80.1	80.1	80.3	79.0	75.6	71.0
		MCDB	80.2	85.5	90.0	95.4	99.0	94.5	87.4	86.4	87.3	86.6	82.9	79.7
	10%	WB	69.4	71.7	76.2	79.6	81.4	80.5	79.4	79.4	79.6	78.0	74.1	69.6
		MCDB	79.6	84.2	89.2	94.7	97.7	92.6	86.6	85.6	86.2	85.2	81.1	78.5
(35) Mean Daily Temperature Range	5% DB	MDBR	23.2	22.9	22.6	21.8	20.5	14.5	10.3	9.3	11.2	16.1	20.5	23.3
		MCDBR	25.0	25.7	25.5	23.9	22.2	18.7	13.4	12.1	14.1	18.5	21.4	24.6
		MCWBR	8.6	7.5	8.2	8.3	7.9	6.2	4.7	4.8	5.2	6.5	7.1	8.6
	5% WB	MCDWR	19.6	21.4	20.9	20.6	20.0	16.3	11.8	10.9	12.3	14.6	15.7	19.0
		MCWBR	8.7	8.1	8.4	8.9	8.5	6.5	5.2	5.1	5.6	6.4	6.6	8.0
(40) Clear-Sky Solar Irradiance	taub		0.481	0.499	0.566	0.593	0.623	0.637	0.589	0.558	0.557	0.540	0.492	0.463
	taud		2.018	1.987	1.845	1.834	1.775	1.780	1.898	1.991	1.987	1.997	2.064	2.085
	Ebn at Noon		251	255	242	237	227	222	233	242	241	240	246	252
	Edh at Noon		52	56	67	69	72	71	63	58	57	55	49	47
(44) All-Sky Solar Radiation	RadAvg		1492	1763	1960	2138	2139	1615	1290	1294	1537	1619	1524	1431
	RadStd		66	64	103	69	65	149	163	129	108	90	93	78

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	+0.45	N/A	-3.01	+1.73	N/A	N/A	N/A	N/A	+252	+250

Nomenclature: See separate page